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From IoT Data to Intelligent Action

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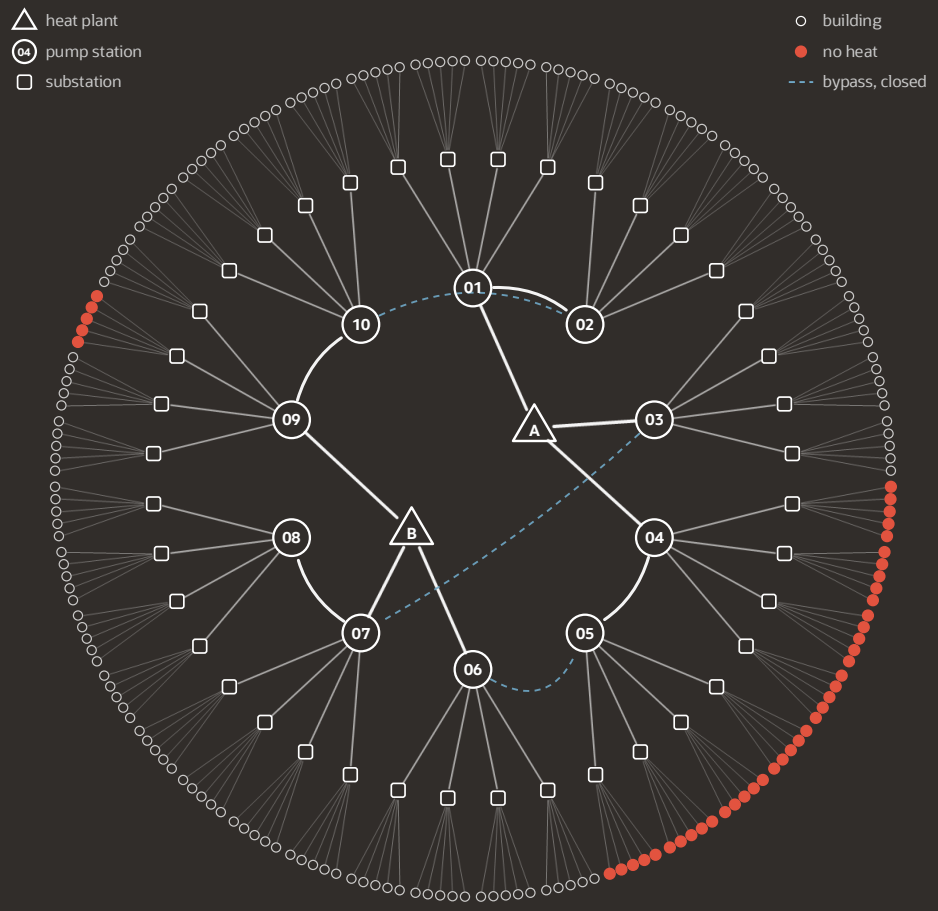


Image by Gemini

Claude
Opus 5.5 High
Anthropic

Monday Morning at 06:40 and −10 °C

45 problems – or two?



Control Room:

- 45 buildings without heat
- 3,160 apartments
- 3 districts
- 1 hospital, 1 nursing home, 1 school

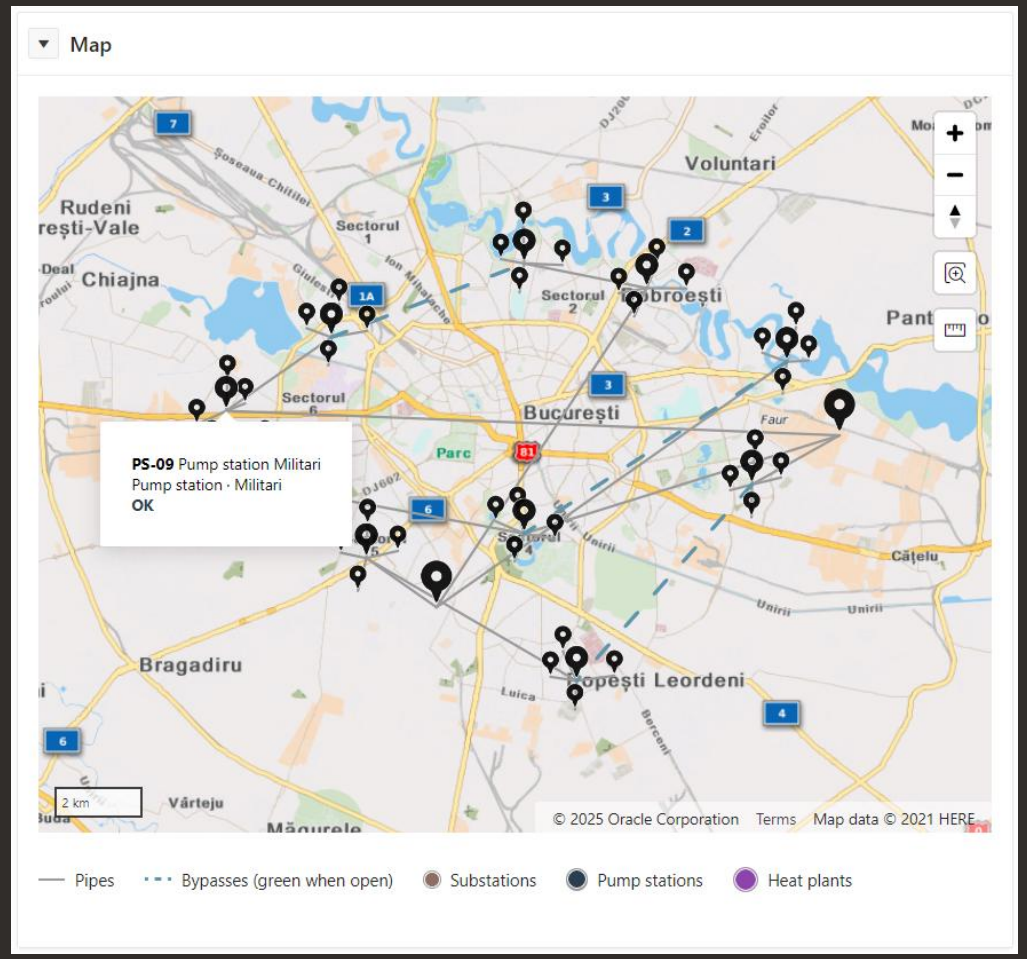


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District heating is a connected system

[BUC] Bucharest / [VIE] Vienna

- [VIE] More than 1,300 km of pipes, about 460,000 households
- [BUC] About 950 km primary and 3,000 km secondary network, the largest in the EU
- [BUC] August 2025: one pipe break, more than 4,000 blocks without hot water
- Every radiator depends on a chain: plant → pump → substation → building
- Sensors everywhere, but alarms arrive one by one



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IoT data is connected data

Relationships matter as much as readings

Relational view

- Readings in rows, dependencies in foreign keys
- "What depends on PS-04?" → recursive SQL
- Fixed depth, hard to read, hard to change
- Topology and events in separate queries

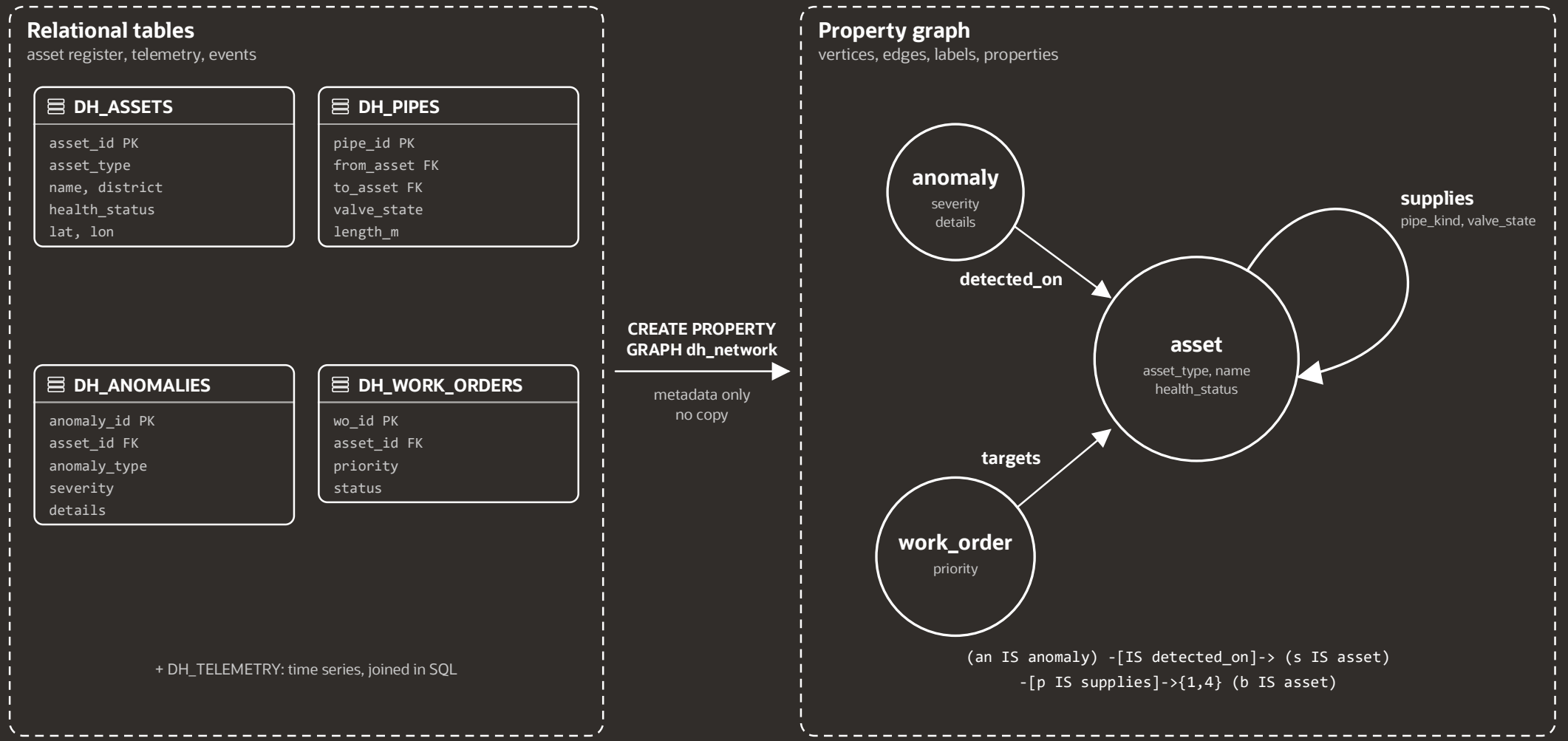
Graph view

- Assets are vertices, pipes are edges
- Paths and patterns are the query
- Events attach to the assets they concern
- Algorithms find importance and reachability

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A graph over the tables you already have

SQL Property Graphs in Oracle AI Database 26ai



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Ask the graph

Patterns, paths, topologies, and SQL in one statement

```
SELECT
  suspect,
  COUNT(DISTINCT building) AS explained
FROM GRAPH_TABLE (
  dh_network
  MATCH (an IS anomaly)
    -[IS detected_on]-> (s IS asset)
    -[p IS supplies]->{1,4}
    (b IS asset WHERE b.health_status = 'ALARM')
  COLUMNS (
    s.asset_id AS suspect,
    b.asset_id AS building
  )
)
GROUP BY suspect
ORDER BY explained DESC
```

- **GRAPH_TABLE** is the table function to return the result set to the outer SELECT
- **MATCH** describes the shape or pattern you are looking for
- **{1,4}**: one (n) to four (m) hops, any depth in between
- Filters inside vertices and edges using **WHERE**
- **COLUMNS** turns matches into rows
- Group, join, aggregate: it's still **SQL**

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Graph algorithms inside the database

DBMS_OGA in Oracle AI Database 26ai

- **PageRank**: which assets matter most
- **Personalized PageRank**: what is closest to these symptoms
- **Weakly connected components (WCC)**: which parts are cut off
- **Bellman-Ford**: shortest weighted paths
- The algorithms return the graph with new properties, read with GRAPH_TABLE

```
SELECT
  asset_id, asset_type,
  ROUND(dist) AS metres_from_plant
FROM GRAPH_TABLE (
  DBMS_OGA.BELLMAN_FORD(
    dh_live_network,
    JSON({'
      "GRAPH_OWNER":"OGBEMEA2026",
      "GRAPH_NAME":"DH_LIVE_NETWORK",
      "ELEM_TABLE":"DH_ASSETS_LIVE_V",
      "KEY_VALUE":{"ASSET_ID":"HP-A"}
    }'),
    PROPERTY(EDGE INPUT length_m DEFAULT ON NULL 0),
    PROPERTY(VERTEX OUTPUT dist)
  )
  MATCH (v IS asset WHERE v.asset_type = 'PUMP_STATION')
  COLUMNS (v.asset_id, v.asset_type, v.dist)
)
ORDER  BY dist;
```


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OCI IoT Platform: device become digital twins

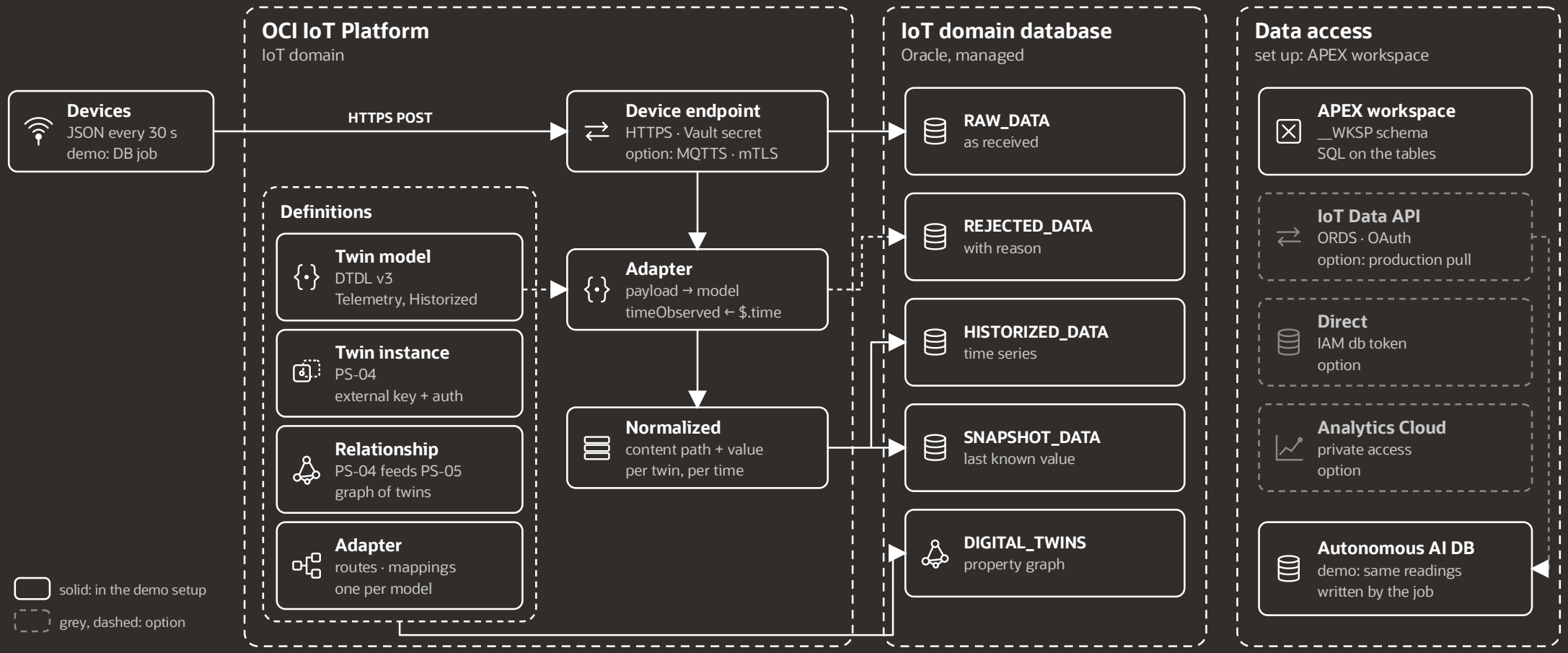
Managed ingestion into an Oracle AI Database

- Model: telemetry and relationships in DTDL v3
- Adapter: maps any JSON payload to the model
- Instance: one twin per device, authenticated by a Vault secret or certificate
- Relationship: twin "feeds" twin – a graph of twins
- HTTPS or MQTTS in; raw, historized and latest values out

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From device to database

Normalized, historized, ready for SQL



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Agents, tools, skills and MCP

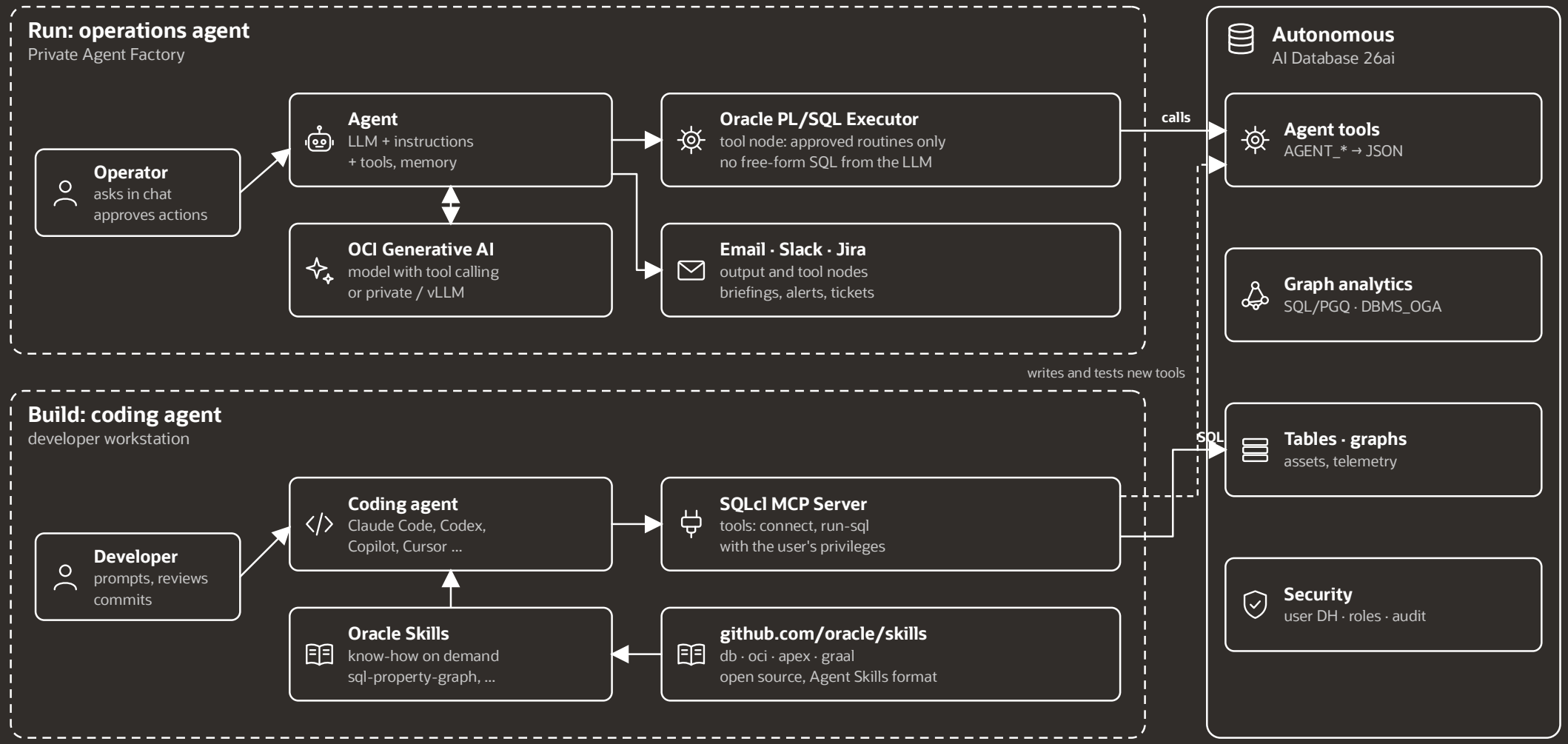
Four terms, one minute

- **Agent**: an LLM with instructions that decides which tools to call
- **Tool**: a function the agent is allowed to call – here, PL/SQL
- **Skill**: know-how an agent loads when a task needs it
- **MCP**: a standard way to connect tools to any agent

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Agentic AI on Oracle: run and build

Operations agents call tools, coding agents write them



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Oracle AI Database Private Agent Factory

Build, run and govern agents next to your data

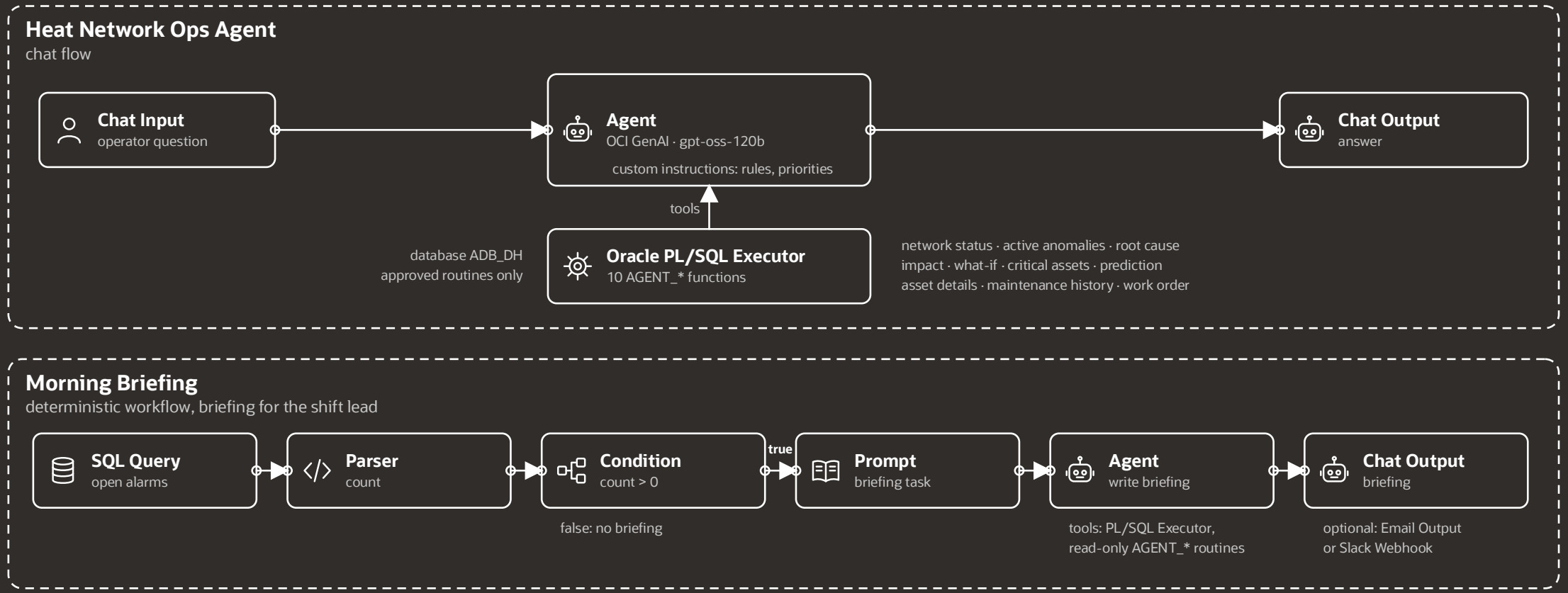
- No-code Agent Builder with flows
- Prebuilt agents: Knowledge, Data Analysis, Deep Data Research
- Runs in containers, in the cloud or on premises
- No extra cost for Oracle AI Database customers

- **Nodes**: Agent, Oracle PL/SQL Executor, SQL Query, MCP Server, REST API
- **Outputs**: chat, email, Slack, Jira
- **LLMs**: OCI Generative AI, private models, others
- **Governance**: approved models, data sources, roles

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Two flows for the control room

A chat agent and a morning briefing



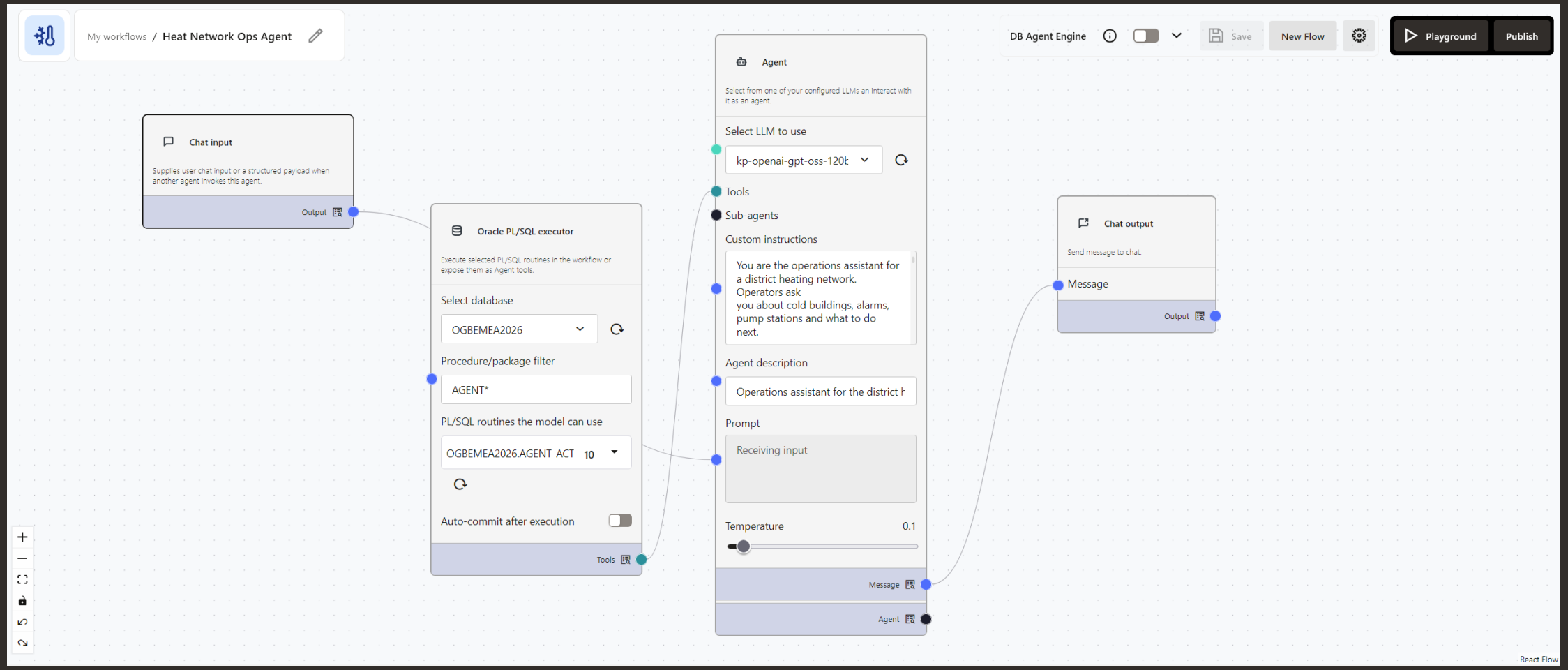
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Ten tools, one database
Approved PL/SQL functions that answer with JSON

Function	Answers	Built with
AGENT_NETWORK_STATUS	What is the situation? Assets by health, buildings and apartments without heat, districts	SQL, JSON
AGENT_ACTIVE_ANOMALIES(min_severity)	Which assets are in alarm or warning, which metric, since when?	SQL, JSON
AGENT_FIND_ROOT_CAUSE(district)	Which upstream asset explains the cold buildings? Equipment problem or sensor fault, with evidence	SQL/PGQ paths, Personalized PageRank
AGENT_IMPACT_ANALYSIS(asset_id)	Who depends on this asset? Critical buildings, bypass options and their effect	SQL/PGQ, what-if
AGENT_WHAT_IF(out_of_service, open_valves)	What if this asset fails and that valve opens?	SQL/PGQ on the live graph, WCC
AGENT_PREDICT_FAILURES(horizon_hours)	What will fail next? Trend, hours to alarm, dependent buildings, last maintenance	Trend anomalies, SQL/PGQ
AGENT_CRITICAL_ASSETS(top)	Which assets matter most for supply?	SQL/PGQ, PageRank
AGENT_ASSET_DETAILS(asset_id)	Latest values, supplier, open work orders	SQL, SQL/PGQ
AGENT_MAINTENANCE_HISTORY(asset_id)	What was done on this asset before?	SQL
AGENT_CREATE_WORK_ORDER(asset_id, priority, title, details)	The only write tool: a work order, after the operator confirms	SQL insert

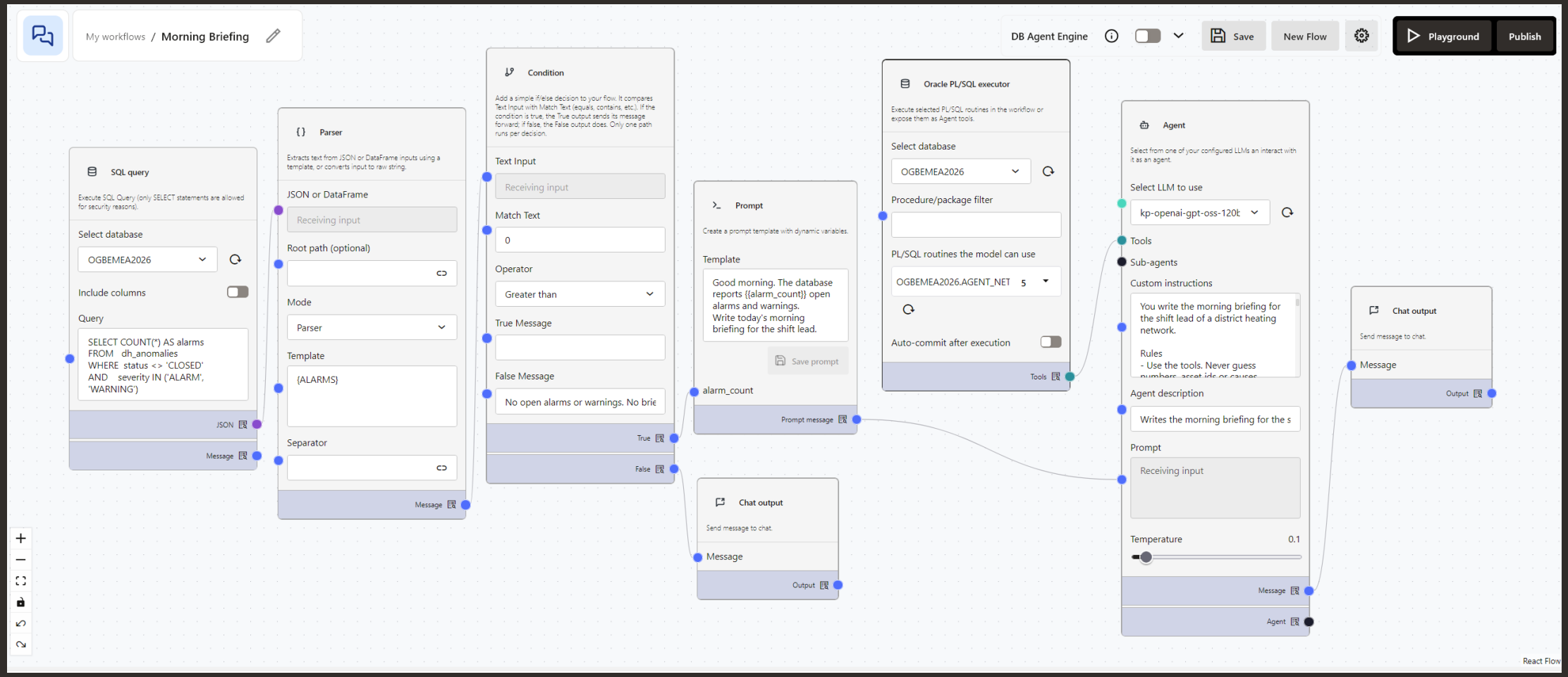
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Heat Network Ops Agent



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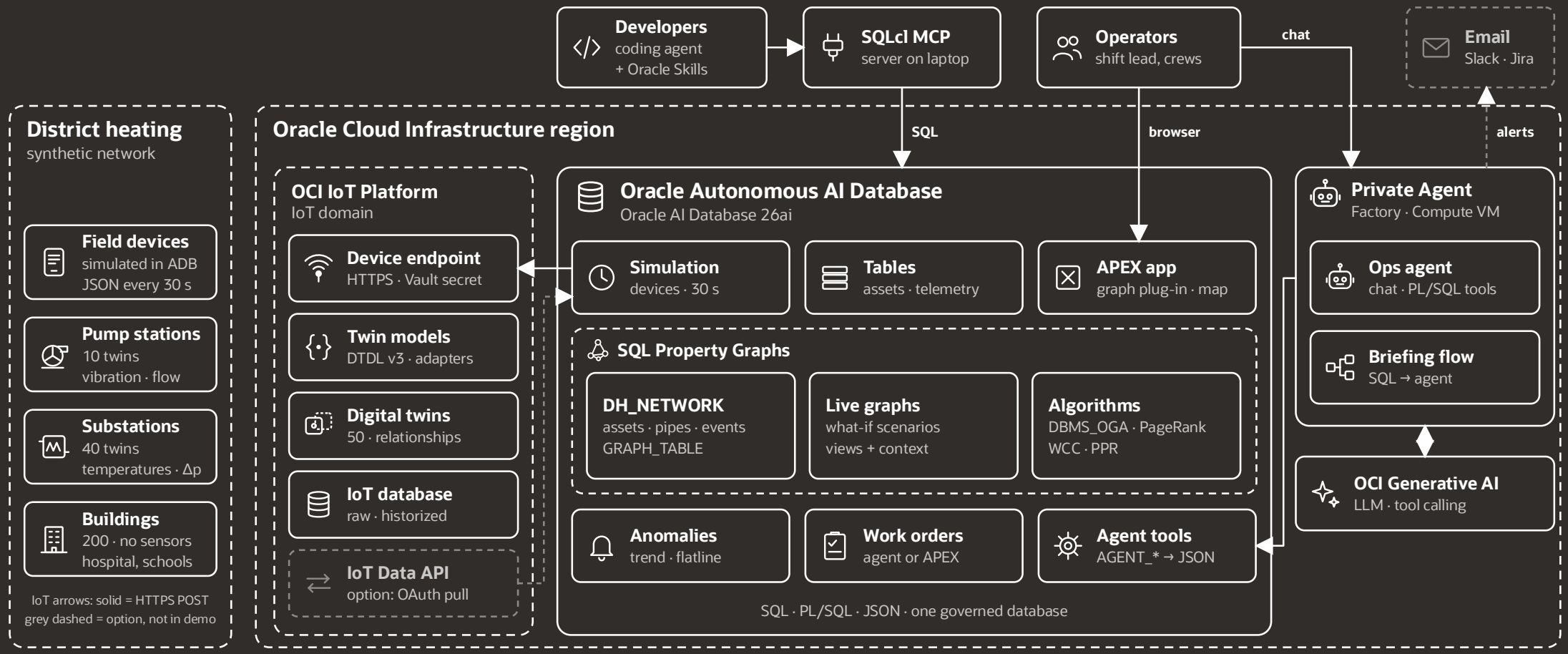
Morning Briefing



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Reference architecture

IoT, graph analytics and agentic AI on one database



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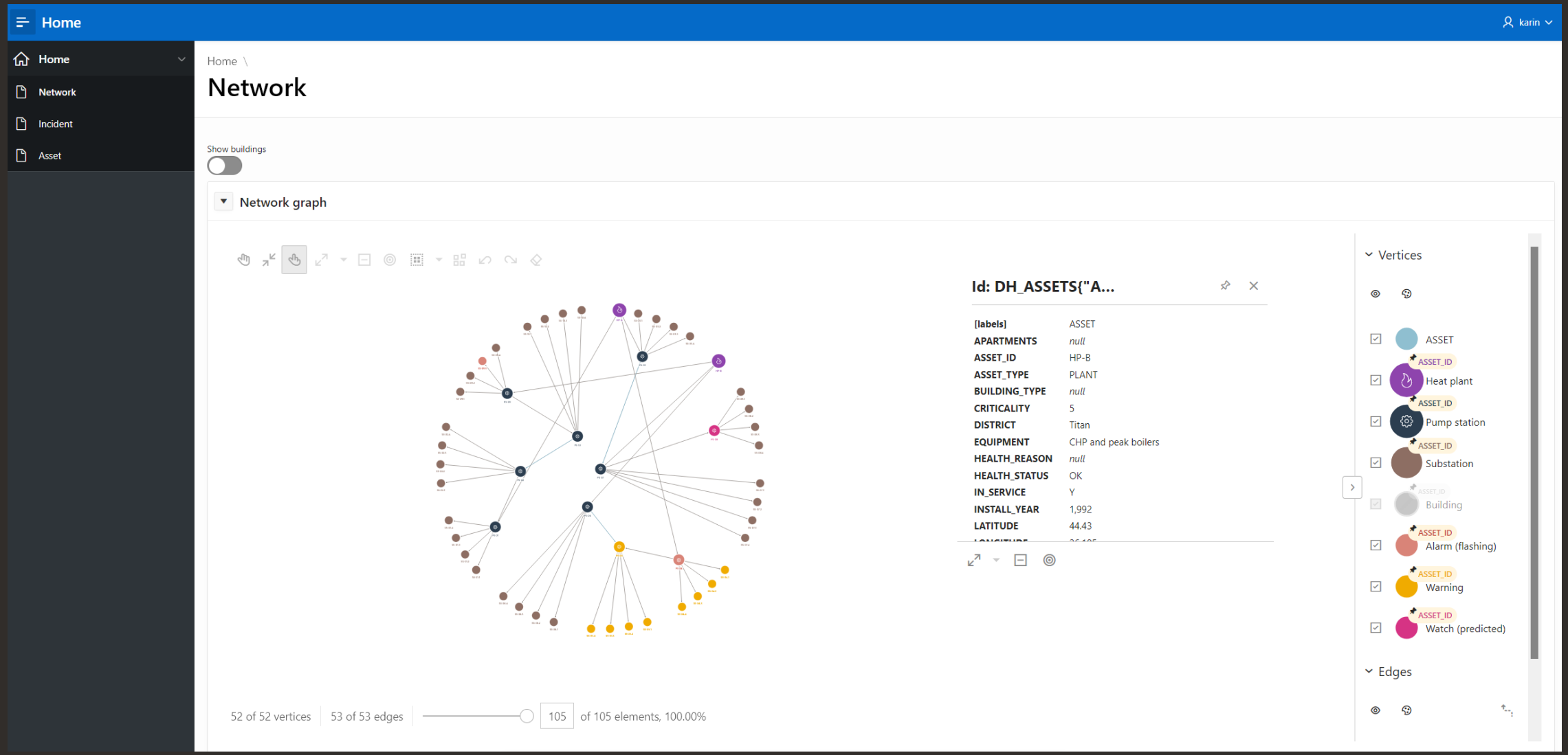
Let's turn the heating back on

Live demo

- ❑ Act 1: IoT data lands in the database
- ❑ Act 2: From symptoms to root cause with SQL/PGQ
- ❑ Act 3: Graph algorithms in the database
- ❑ Act 4: Graph apps with APEX
- ❑ Act 5: The agent recommends, the operator acts

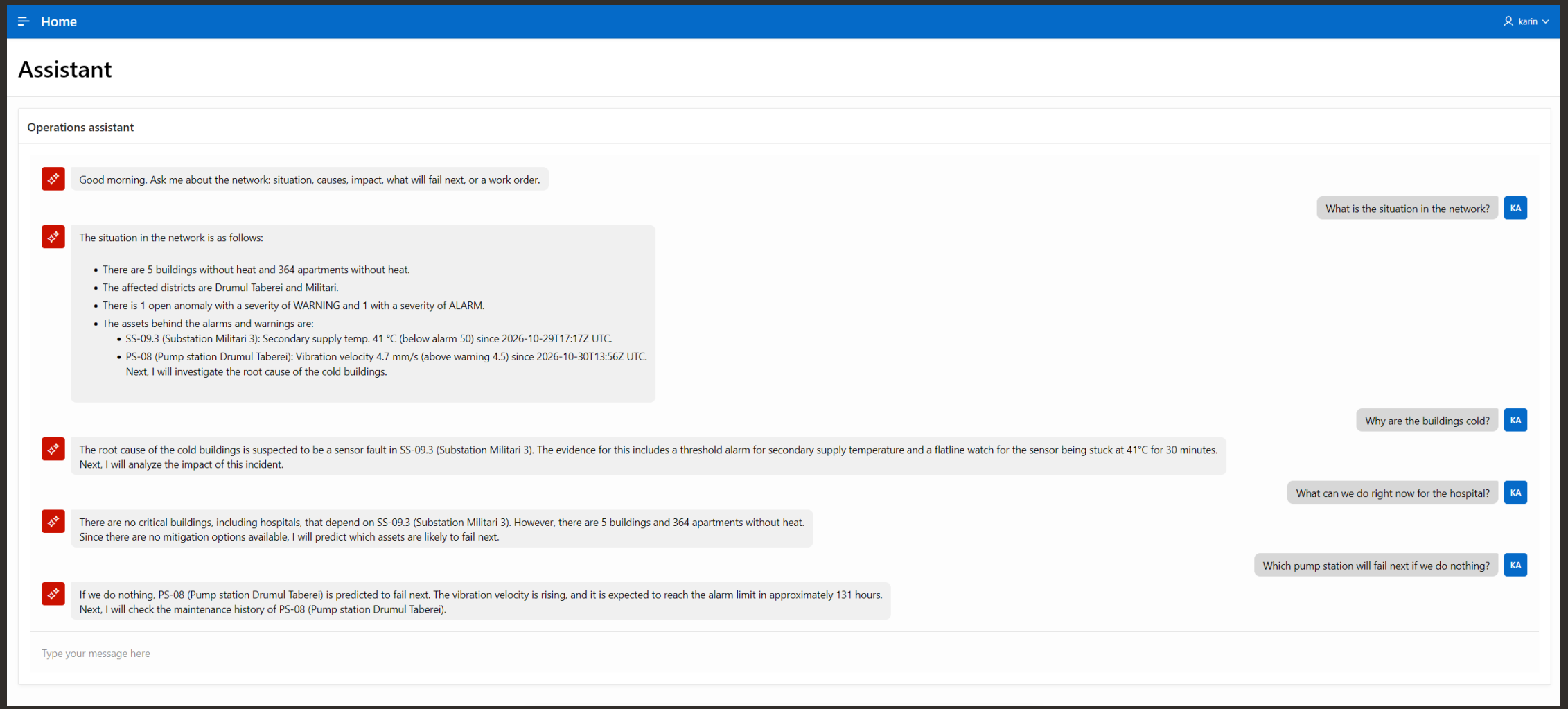
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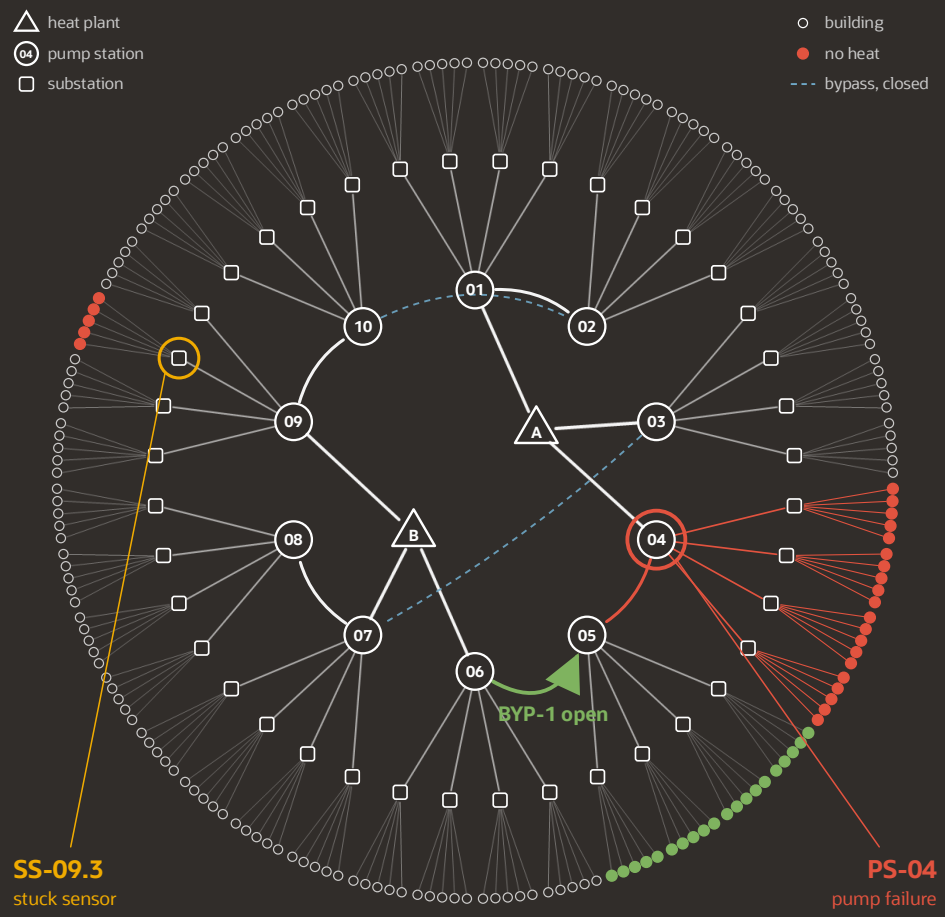
Asking the Heat Nework Ops Agent



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From IoT data to intelligent action

Three things to take home



1. Model the network as a graph on the tables you have
2. Let patterns and algorithms find causes and impact
3. Give agents governed tools, not database access

- Bonus (as of tomorrow): github.com/karinpatenge/events/2026/11_RoOUG_AOUG

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Thank you – questions?



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Abbreviations and names (1/2)

- A2A: Agent2Agent, an open protocol for agents that call other agents
- ADB: Oracle Autonomous AI Database
- AI: artificial intelligence
- APEX: Oracle APEX, the low-code platform inside the Oracle database
- CTE: common table expression, the WITH clause (recursive: it refers to itself)
- DBMS_OGA: Oracle Graph Algorithms, a PL/SQL package in Oracle AI Database 26ai
- DTDL: Digital Twins Definition Language, JSON-based, version 3
- EU: European Union
- HTTPS: encrypted HTTP
- IoT: Internet of Things
- ISO: International Organization for Standardization
- JSON: JavaScript Object Notation
- LLM: large language model
- MCP: Model Context Protocol, a standard that connects tools to agents
- MQTT / MQTTS: a lightweight messaging protocol for devices; MQTTS is the encrypted version
- MSET-SPRT: Multivariate State Estimation Technique – Sequential Probability Ratio Test, an anomaly detection algorithm in OML
- OCI: Oracle Cloud Infrastructure
- OML: Oracle Machine Learning
- ORDS: Oracle REST Data Services
- PGQ: property graph queries. SQL/PGQ is part 16 of the ISO SQL standard (SQL:2023)
- PL/SQL: Oracle's procedural language for SQL
- PPR: personalized PageRank
- QR code: quick response code
- R^2: coefficient of determination, how well a straight line fits (1 = perfectly)
- REST: representational state transfer, a style of web API
- SQL: Structured Query Language
- SQLcl: SQL Developer command line
- VM: virtual machine
- WCC: weakly connected components

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Abreviation and names (2/2)

- DH: district heating, the prefix of all demo objects (DH_NETWORK, DH_TELEMETRY, …)
- HP-A, HP-B: heat plants
- PS-01 … PS-10: pump stations
- SS-04.1: substation 1 below PS-04. It transfers heat from the network to the buildings' own circuit.
- B-04.1.2: building 2 at substation SS-04.1
- P-0003: pipe
- BYP-1: bypass pipe with a valve, normally closed, from PS-06 to PS-05
- t_sec: secondary supply temperature, the water going to the building
- t_supply / t_return: primary supply / return temperature, network side
- dp: differential pressure
- bearing_tmp: pump bearing temperature
- Units: mm/s (vibration velocity), °C, bar, kW, m³/h

- Anomaly types:
  - THRESHOLD: a limit is crossed
  - ZSCORE: far from the 7-day mean
  - TREND: heading for a limit
  - FLATLINE: a constant value, possibly a stuck sensor
  - NO_DATA: the twin is silent
- Severity:
  - WATCH < WARNING < ALARM
- Work order priority:
  - P1 (highest) to P3
- Graphs:
  - DH_NETWORK: assets, pipes, anomalies and work orders
  - DH_LIVE_NETWORK: the same network with what-if scenarios
  - DH_DEPENDENCY: edges reversed, pointing upstream

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Q&A

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